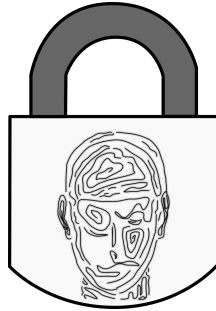


FraudLock: Developing Robust Deepfake Detection Methods for Face Recognition



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Chapter 1

Introduction

Deepfake generative AI technology provides new opportunities to create rich and high-quality content in multimedia applications such as virtual reality[1], film production[2], and neural presentation[3, 4]. However, its misuse has become a major threat to society and has led to problems such as fraud, scams, and extortion.

1.1 Problem Domain

The advancement in deepfake technology poses a threat to a person's identity. Some existing detection methods detect deepfake but they mainly focus on isolated features, such as lips or eyes. However, they often overlook the correlation between the facial muscles, jaw, and lip movements, leading to increased false positives and a lack of reliability in real-world applications. Our goal is to develop a robust deepfake detection system with a multi-modal approach that synchronizes the movement of the eye, lip, audio, and cheeks muscles. This research aims to provide a cross-platform optimized app for both IOS and Android mobile devices.

1.2 Research Problem Statement

The advancement in deepfake technology poses a threat to a person's identity. Existing detection methods, which rely on visual artifacts, audio-visual inconsistencies, or biometric features, fail to capture the complex manipulations found in advanced deepfakes, particularly the interaction of lip motions and facial muscle dynamics. Existing deepfake methods often overlook the correlation between the facial muscles, jaw, and lip movements, leading to increased false positives and a lack of reliability in real-world applications. Our goal is to develop a robust deepfake detection system with a multi-modal approach that synchronizes the movement of the eye, lip, audio, and jaw muscles. This research aims to provide a cross-platform optimized app

for both IOS and Android mobile devices.

- How can a multitask learning model effectively detect lip-audio synchronization in deep-fake videos?
- What impact do lip movements have on cheek muscle dynamics, and how can this be modeled for deepfake detection?

Chapter 2

Literature Review

2.1 Related Research

2.1.1 FAMM: Facial Muscle Motions for Detecting Compressed Deepfake Videos Over Social Networks

2.1.1.1 Summary of the Research Item

Paper "FAMM: Facial Muscle Motions for Detecting Compressed Deepfake Videos Over Social Networks"[5] addresses the menace of Deepfake videos, particularly those disseminated over social networks wherein video compression is inescapable. The authors have advocated a framework called FAMM (Facial-Muscle-Motions-Based), which improves the detection of compressed Deepfake videos. The methodology consists of three modules: facial landmark extraction, facial muscle motion feature construction, and probability prediction fusion.

Firstly, for a particular video, the pipeline picks facial landmarks from consecutive frames through suitable face alignment techniques and DLIB. These facial landmarks then form geometric features which capture the muscle movement in five different sensory areas along with the entire face. The temporal dynamics of these movements then create patterns for detection as unnatural, thus suggesting Deepfakes by FAMM. A final detection is developed based on Dempster-Shafer theory by fusing the probability scores from both GRU and SVM classifiers to ensure more robustness to compression artifacts. Extensive experiments demonstrate the superiority of FAMM in compressed Deepfake video detections, holding even accuracy in released Facebook and YouTube videos.

2.1.1.2 Critical Analysis of the Research Item

The FAMM framework has several advantages that make the above solution effective for Deepfake detection. First, the application of facial muscle motion features is quite unique for giving a geometric view beyond the conventional artifact-based detection methods. It is based on the continuity of natural facial movements, which works to fight such noise commonly seen with compression in videos found on social networks. Deep feature construction, uses both the distance and angle metrics over different facial regions, increasing the model's ability to counter subtle manipulations.

Further, Dempster-Shafer theory for probability fusion gives a good integration of the strengths of GRU and SVM classifiers to improve the overall detection accuracy. Further, the theoretical analysis on mutual information and invariance of facial landmarks even after compression adds on to the creditworthiness of the robustness of the framework. However, the work also has some drawbacks. The method based on precise facial landmark extraction sometimes faces the problem of low-resolution or heavily occluded videos that can affect the feature accuracy. More than that, The superior performance on standard datasets demonstrates the work well; however, its adaptability to diverse and evolving Deepfake generation techniques has not been adequately evaluated in this work. The computational complexity of processing continuous facial landmarks and temporal features can further affect the scalability of this model, especially for real-time applications on large-scale social networks.

Overall, FAMM holds promising potential in the area of Deepfake detection, especially with regards to compressed videos. Future work can improve the robustness of the framework for landmark extraction, and develop its adaptability to more techniques for making the model rock-solid.

2.1.1.3 Relationship to the Proposed Research Work

Our proposed method is the extension of FAMM framework, taking visual and audio data to further improve Deepfake detection. Focusing only on facial muscle movements in compressed videos, our method also includes lip-audio synchronization analysis for cheek muscles. We can better detect mismatches in lip movements with the audio by aligning features of audio with frames of the video. Our use of attention mechanisms supports better focusing of the model on key areas, which improves the accuracy of detection.

Unlike the approach adopted in the FAMM, where the Dempster-Shafer theory combines GRU and SVM classifiers, the present proposal adopts MLP with feature fusion techniques in order to combine the predictions from different modules in a way of simplifying the architecture while offering an effective combination of different feature sets.

2.1.2 Preventing DeepFake Attacks on Speaker Authentication by Dynamic Lip Movement Analysis

2.1.2.1 Summary of the research item

This paper talks about the vulnerability of lip-based speaker authentication systems to DeepFake attacks by focusing on the analysis of dynamic lip movements rather than by relying on static features that are vulnerable to manipulation. In their work titled "Preventing DeepFake Attacks on Speaker Authentication by Dynamic Lip Movement Analysis," [6] the authors propose a visual authentication scheme using a Deep Convolutional Neural Network (DCNN). A proposed model called SA-DTH-Net is divided into two major parts: the Fundamental Feature Extraction Network (FFE-Net) for dynamic lip motion analysis and the Representative Lip Feature Extraction and Classification Network (RC-Net) for the purpose of extracting high-level features. It focuses more on speaker authentication via liveness through dynamic lip movements, thus reducing the threat from static lip features vulnerable to DeepFake technology.

The model clearly demonstrated excellent performance on the GRID and MOBIO datasets, thereby outperforming the authentication accuracy of three state-of-the-art visual speaker authentication algorithms against human imposters. Another strength of the work was on its robustness against DeepFake-generated attacks and ensured that the usage of multitask learning within the RC-Net contributes to reliable DeepFake detection by optimizing the network for both feature reconstruction and classification accuracy. Importantly, this proposed method does not rely on knowledge of specific DeepFake spoofing techniques, thus is flexible with differing spoofing strategies.

2.1.2.2 Critical Analysis of the Research Item

In a principal strength, the paper introduces a novel architecture in SA-DTH-Net which integrates dynamic lip movement analysis along with the implementation of robust feature extraction mechanisms. By prioritizing temporal dynamics over static features, the described method complicates liveness detection and makes the authentication process even more difficult to deceive with DeepFake technologies. The addition of the multitask learning within RC-Net further strengthens the model because of the high discriminative ability of the feature extracted for authentication purposes, but also representative of individual talking styles, which enhances the system's resilience against impersonation attacks.

However, the study has certain limitations. Mostly, reliance on prerecorded data may impede the model's performance when applied to increasingly diversified and variable datasets that are often faced during real-world scenarios. There could also exist variability in real-world environment settings-lighting, camera angles, and occlusions introduced to the system that may impact the accuracy of authentication. The computational cost of this presented network, mainly its training, has the potential to affect the deploymentability of the system to resource-

poor settings. On the other hand, the reliance on precomputed optical flow maps is effective in motion-capture applications, but it may become an disadvantage where optical flow estimates are noisy or poorly valued and degrade overall authentication performance.

2.1.2.3 Relationship to the proposed research work

Where the SA-DTH-Net explores the use of dynamic lip movements for authentication of speakers, our method makes use of both lip-audio synchronization and cheek muscles movement analysis to produce a richer test for video authenticity.

Multitask learning is used for optimizing our model to multiple objectives like SA-DTH-Net. This would further augment its ability in effectively detecting DeepFake manipulation.

However, our solution will be a bit different from SA-DTH-Net, as it mainly focuses on speaker verification. This method detects DeepFakes by looking at lip motion and audio signal mismatches, and also cheek muscle mismatch. Our model can better realistically identify unnatural patterns of manipulations suggesting DeepFake maneuvers since it aligns audio features with the corresponding video frames and allows use of attention mechanisms. Instead, whereas SA-DTH-Net employs multitask learning within its RC-Net for the reconstruction of features and classification, our proposed solution utilizes Multi-Layer Perceptrons and feature fusion techniques to make predictions based on various modules. This integrates multiple feature sets in a streamlined fashion and enhances the accuracy and robustness of the detection of forged videos.

2.1.3 Improving the Efficiency and Robustness of Deepfakes Detection through Precise Geometric Features

2.1.3.1 Summary of the research item

In the paper "Improving the Efficiency and Robustness of Deepfake Detection through Precise Geometric Features" [7] the authors propose a unique detection framework that uses facial landmarks and a two-stream Recurrent Neural Network to enhance efficiency and robustness in detecting deepfakes. This research includes three key steps in their methodology: **Face Preprocessing:** In this step the faces are cropped from the video and aligned it using affine transformation. **Landmark calibration:** In this process they use the Lucas-Kanade optical flow algorithm which predicts landmark movement of the next frame by learning the movements from the previous frames, and a Kalman filter which integrates both the detected and LK prediction results to compute more stable landmark positions. It used Kalman gain to decide how much to trust the predictions versus the detection result based on the relative variance to decide when the detection is more unstable than the prediction finally, **Feature extraction and**

classification: After getting the calibrated landmarks these landmarks are embedded into two types of feature vector one for facial shape α and other one for movements between the frames β and then these two feature vectors are feed into two separate RNNs g_1 and g_2 each RNN produces its prediction and then these predictions are averaged to produce the final classification result.

The authors conclude that this research had a sound approach in the focus on geometric features rather than full image, pixel-level analysis increases the computational cost. They come out showing the model referred to as Landmark Recurrent Network (LRNet), which performs effectively by bringing forth the union of both feature vectors of facial shape and the temporal movements, making it a more efficient and flexible solution for deepfake detection compared to conventional pixel-level approaches.

2.1.3.2 Critical analysis of the research item

One of the strong points of this approach is the ability to represent features efficiently. In this research, the investigator uses only facial landmarks, rather than whole images which is a very efficient way of deepfake detection. The method is also simplified as it concentrates only on the geometry of the face eliminating unnecessary complexities associated with analyzing images on a pixel level. In addition, a Landmark Calibration module using the Lucas-Kanade optical flow method and Kalman filter is added to deal with the problem of very jittery landmarks that lead to noise and inaccurate results. The LK algorithm tracks movement and the Kalman filter reduces the noise introduced by jittery landmarks. Another advantage is that it employs a Two Stream RNN in which one stream accommodates the face shape while the other stream takes the image frames over a period of time. This decision improves the system's ability to discriminate between inconsistencies in the structure of the face and its motion, which is very important for deepfake detection. In addition, the method is quite flexible and comes with wide applicability as it does not rely on any particular technique of landmarking which enhances its generalization and adaptability to different datasets.

Nonetheless, this methodology has certain limitations. Due to the forward-backward check of the previous and next frames the use of the Kalman filter may improve the accuracy and reduce the noise but it introduces additional complexity and computational overhead. Continuous forward and backward checks, slow down the real-time detection. The selection of parameters like inherent variance Q and approximate relative variance D_r is empirical. It may work well for controlled experiments but may not generalize to real-world situations where factors like lighting could lead to inaccurate results and variances are difficult to predict or highly variable.

2.1.3.3 Relationship to the proposed research work

While the referenced study focuses on using facial landmarks and a two-stream Recurrent Neural Network (RNN) to detect Deepfakes, our method extends this by incorporating both visual and audio data streams. By integrating lip-audio synchronization and cheek muscle movement analysis, we enhance the detection capabilities beyond geometric features alone.

Additionally, similar to the use of multitask learning in the reviewed framework, our approach employs Multi-Layer Perceptrons (MLPs) and feature fusion techniques to combine predictions from different modules, improving overall accuracy and robustness. This extension allows our model to better identify inconsistencies and unnatural patterns in both facial movements and audio signals, providing a more comprehensive solution for Deepfake detection.

2.1.4 DF-TransFusion: Multimodal Deepfake Detection via Lip-Audio Cross-Attention and Facial Self-Attention

2.1.4.1 Summary of the research item

In the paper "DF-TransFusion: Multimodal Deepfake Detection via Lip-Audio Cross-Attention and Facial Self-Attention"[8], the authors propose an innovative deepfake detection system that utilizes deep learning techniques to analyze both audio and video data. The architecture employs a modified VGG-16 feature extractor for video data while audio data and lip regions are processed in parallel using an audio transformer encoder. Face and lip regions of the videos are very accurately obtained with the help of MTCNN which acts as an adjunct for the region of interest extraction process. They used tubelet embedding method which captures spatial and temporal relationships between frames at the same time, unlike the uniform frame sampling approach, where each 2D frame is tokenized separately and temporal embedding information needs to be provided independently to the transformer encoder. This embedding strategy enables to promotion of the efficiency of the model by making it much easier to perform cross-attention between lip movement and audio by making it easier for the model to tell whether the audio and the lips are properly synchronized or not which is one of the features that deepfakes tend to alter. Finally, the outputs from both the audio and video transformer encoder modules were then passed through the head of a multilayer perceptron (MLP) to generate the final classification label.

The study concludes that this multimodal approach outperforms models that analyze modalities independently. They highlight the potential of multimodal analysis in deepfake detection. It emphasizes the need for continuous adoption of these models to newer deepfake generation methods.

2.1.4.2 Critical analysis of the research item

The strengths of this research are notable. First, the use of VGG-16 for feature extraction and a transformer encoder with self-attention and cross-attention mechanisms, which are highly effective for handling sequential data such as video and audio. This combination seems well-chosen, as VGG-16 is known for its efficient performance in extracting features from images, while the transformer model captures both spatial features through self-attention and temporal relationships through cross-attention, making it suitable for tasks like lip-audio synchronization. Another strength is the use of cross-attention for detecting lip-audio synchronization inconsistencies, a powerful indicator of manipulation since deepfakes often fail to synchronize these two modalities.

Additionally, this research improves efficiency in processing videos by reducing the number of frames from 300 to 30 in the video-only pipeline, as deepfakes typically persist throughout the entire video. This approach helps to address computational costs. The audio-visual data fusion using a multi-layer perceptron (MLP) also improves the model-centric manipulation detection capability against attacks that affect only one of the modalities. Also, with cuboid embeddings, since both space and time are fused simultaneously, spatiotemporal correlation is captured more efficiently making the attention mechanisms easier to implement and learn these correlations better.

Nonetheless, there are some potential drawbacks in this strategy. To begin with, the lip region has been pre-processed where frames have converted into grey-scaled images. This may cause loss of useful information which may involve lip movements or texture around the mouth region, which helps in the deepfake detection. Color pictures versus grayscale inputs have not been tested or the results compared by the author, therefore the justification for this pre-processing step is insufficient. In addition, the study does not address the model's false positive or false negative rates, thus making it challenging to ascertain what deepfakes would confuse the model the most.

Another concern is the conversion of the audio signal from stereo to mono to reduce input size. While this decreases computational complexity, it risks losing spatial audio cues, potentially masking manipulations, especially if the deepfake is applied only to one channel of the audio. Lastly, while the use of VGG-16 and two separate transformer encoders for the audio and video pipelines is effective, it significantly increases computational complexity, which may hinder the approach's applicability for real-time detection. More efficient architectures may be necessary for such applications.

2.1.4.3 Relationship to the proposed research work

DF-TransFusion uses lip-audio cross-attention and facial self-attention to identify synchronization issues, our method extends this by also analyzing cheek muscle movements alongside lip

movements.

Like DF-TransFusion, we employ multi-task learning to enhance detection capabilities. Additionally, we align audio features with video frames and use attention mechanisms to focus on both lip and cheek regions, allowing our model to detect a broader range of inconsistencies. Instead of relying solely on transformer encoders, our approach utilizes Convolutional Neural Networks (CNNs) and Multi-Layer Perceptrons (MLPs) for feature extraction and classification, facilitating more effective feature fusion.

2.1.5 Lips Are Lying- Spotting the Temporal Inconsistency between Audio and Visual in Lip-Syncing DeepFakes

2.1.5.1 Summary of the research item

In the paper "Lips Are Lying: Spotting the Temporal Inconsistency between Audio and Visual in Lip-Syncing DeepFakes"[9], the authors introduce LipFD, a novel framework designed to detect lip-sync forgeries by analyzing discrepancies between audio and video in deepfake technologies. The main goal of the LipFD framework is to provide a solution to the challenge of detecting lip synchronization errors in input videos that have been manipulated. The architecture includes a Global Feature Encoder that incorporates self-attention to realize long audio-visual relationships so as to address lip sync problems.

LipFD is made of three major modules: a Region Awareness module, which provides feature weighting, hence enabling the model to concentrate on the most important parts of input. Also, the architecture uses crop encoding that looks at wider portions of the input in order to carry out lip motion more effectively. The authors implement LipFD testing against others' leading methods and demonstrate that it achieves better performance across multiple datasets that include the AVLips proposed dataset, FF++ and DFDC. Metrics such as the accuracy (ACC), average precision (AP), false positive rate (FPR), and false negative rate (FNR) are utilized to evaluate the efficiency of the model. Last but not least, the resilience of LipFD is evaluated against popular perturbation attacks and is found to outperform contemporary deepfake detection techniques.

2.1.5.2 Critical analysis of the research item

The architecture in the article explains that the Global Feature Encoder uses self-attention mechanisms which are appropriate for challenging tasks such as lip-sync picture which entails the notion of temporality across several image frames within the suspected video. In this way, the architecture allows for the model to easily understand the long-range relationship that exists between lip movements and audio signals which would help during the detection of small

details in contrast to deep fake videos. Thanks to the self-attention mechanism, the model has the ability to attend to specific features across the various frames, which could increase the degree of manipulation that can be detected, given that many existing approaches tend to be destructive.

One important aspect of the proposed architecture is that an MLP classifier using a Binary Cross-Entropy loss function is employed. This is a common approach for other classification tasks, but the authors would improve their presentation if they explained how various hyperparameter choices such as the learning rate or layer sizes affect the training process and performance of the model. For example, incorporating specific values or ranges for these hyperparameters would allow readers to understand their impact on convergence speed and accuracy.

The loss function, L_{RA} , mathematically defined as

$$L_{RA}(\theta_{GR}, \theta_{RA}) = \sum_{j=1}^N \sum_{i=1}^T \exp(\omega_i^{max} - \omega_i^h),$$

emphasizes the importance of the most discriminative features in lip-sync detection. This focus on forged areas within the frames improves detection accuracy, as the model becomes particularly attuned to detecting anomalies in lip movements. However, further discussion on the implications of varying the steepness of the loss curve by adjusting hyperparameters, such as k , would deepen the analysis of how this affects the model’s sensitivity.

The evaluation metrics, including accuracy (ACC), average precision (AP), false positive rate (FPR), and false negative rate (FNR), provide a thorough assessment of the model’s performance. A noticeable achievement is the 3.8% accuracy gain over RealForensics reported across three deepfake datasets. Nonetheless, the addition of more metrics, F1 and ROC-AUC in particular, would help assess the performance level in more detail, especially when it comes to separating slight alterations. Within current results, the model is capable of demonstrating an average precision of 90% which is impressive and indicates a high capability in spotting altered images; however, what these results imply when applied to the practical realm needs to be addressed.

The exploration of the generalization of the model is particularly interesting as it proves the flexibility of the architecture with respect to new data produced by different methods including MakeItTalk. This suggests robustness, but it would be useful to analyze the impact of such factors as the lack of use of certain manipulations in the training data set on the scope of the model.

In addition, while performing the robustness analysis, it was shown that the model copes well with standard perturbations even without decreasing the performance in relation to a number of adjustments, which is better than RealForensics performs for brightness and contrast perturbations, and similar to that of the competitors for other types of distortions. The paper also states that near LipFD accuracy is kept above 85% under these perturbations, but more interesting

would be if current metrics are sensitive to those perturbations how performance on each of the metrics changes with respect to the other one. To further increase robustness, recommendations on how the model should be altered or what type of data augmentation strategies to implement could be useful.

Even though these advantages of this strategy must be acknowledged, some drawbacks present themselves. The preprocessing step of converting lip region frames to black and white may inadvertently strip critical texture and movement information, which could be vital for accurately detecting deepfakes. The lack of exploration of the impact of using color images versus grayscale inputs weakens the rationale behind this preprocessing choice. Additionally, converting audio signals from stereo to mono raises concerns regarding the loss of spatial audio cues that may be crucial in identifying manipulations, especially if the deepfake modifies only one audio channel.

The architecture, while effective in leveraging self-attention and MLP classifiers, may also introduce significant computational complexity, which could hinder its applicability for real-time detection scenarios. This complexity raises questions about scalability and efficiency, suggesting that exploring alternative architectures or techniques, such as pruning or quantization, might be beneficial for deployment in practical applications.

2.1.5.3 Relationship to the proposed research work

LipFD effectively detects temporal inconsistencies between audio and lip movements using self-attention mechanisms, our method incorporates additional facial features and employs multi-task learning to enhance detection accuracy. This comprehensive analysis allows our model to identify a wider range of Deepfake manipulations, improving robustness and reliability in various video conditions.

2.1.6 Mobile DeepFake Detection through Machine Learning-based Corneal-Specular Backscattering

2.1.6.1 Summary of the research item

The research item under review introduces MobiDeep[10], real-time DeepFake detection mobile application designed to operate without relying on cloud services. The authors propose a machine learning-based solution that leverages corneal-specular backscatter (CSB) images of human eyes. MobiDeep hypothesizes that existing DeepFake generation techniques struggle to accurately mimic the reflective elements of the eyes, particularly under varying environmental conditions. By focusing on the most reflective area of the face—the eyes—MobiDeep extracts features related to color components, shapes, and textures of CSB images, alongside Facial

Image Environmental Parameters (FIEP) such as illumination and background settings.

The application employs lightweight Deep Neural Network (DNN) architectures, including MobileNet-V2, EfficientNet-B0, and DenseNet-121, to achieve high accuracy and rapid detection speeds. Experimental results demonstrate that MobiDeep, particularly when utilizing MobileNet-V2, attains an impressive accuracy of 98.7% (custom dataset) with detection times under 200 milliseconds on popular mobile devices like Samsung Galaxy S9 and iPhone 11. Additionally, the creation of the MobiDeep-DFD dataset, comprising over 4,000 annotated facial images, underpins the robustness of the proposed detection methodology.

2.1.6.2 Critical analysis of the research item

The research on MobiDeep demonstrates several strengths, including its innovative use of corneal-specular backscatter images to enhance DeepFake detection accuracy, the development of a real-time, cloudless mobile application that ensures user privacy and broad platform compatibility, and impressive performance metrics with a 98.7% accuracy rate and detection speeds under 200 milliseconds on common mobile devices.

Additionally, the creation of the comprehensive MobiDeep-DFD dataset with diverse annotations supports the model's ability to generalize across various lighting conditions and facial variations. On the contrary, the research carries some drawbacks as well, for instance, the narrowness of concentrating only on the CSB images, which likely will be outdone by more recent DeepFake applications, and the need for uniform environmental parameters, which may affect precision in differing conditions.

Moreover, the dataset may be homogenous because it may not cover many types of ethnicities, older persons or young children, or accessories worn on the faces of the persons captured limiting the performance in different situations. The analysis presented in the paper is also incomplete as it does not include extensive comparison with various modern DeepFake detection approaches, besides the mobile application may have limited resources when run on older or less powerful smart-devices making it less accessible and ineffective. To conclude, MobiDeep's approach to detection of DeepFakes on mobile devices is novel and appears to be highly promising. However, these issues need to be resolved for the enhance resilience and wider reach of the solution.

2.1.6.3 Relationship to the proposed research work

In addition to employing these detection techniques, we have also implemented them within an application designed to facilitate the user experience across several mobile operating systems. The enhancement focuses on accessibility and usability so that even those with moderate devices will still enjoy the full benefit of robust DeepFake detection mechanisms on their phones. We also extend the analysis into the multi-modal data realm by employing audio-visual syn-

chronization techniques with CSB image attributes to enhance the detection performance and reliability in various environments and across various device types.

2.2 Analysis Summary of research items

Paper	Summary of Critical Analysis
FAMM: Facial Muscle Motions for Detecting Compressed Deepfake Videos	The FAMM framework effectively employs facial muscle motion features, offering a geometric perspective that counters compression-induced noise. It combines GRU and SVM classifiers via Dempster-Shafer theory to enhance accuracy. Limitations include reliance on precise facial landmark extraction, which may falter in low-resolution scenarios, and concerns over adaptability to evolving deepfake techniques. Scalability for real-time applications also poses challenges.
Preventing DeepFake Attacks on Speaker Authentication by Dynamic Lip Movement Analysis	The SA-DTH-Net architecture is a key strength of this study, focusing on dynamic lip movement analysis to enhance liveness detection, making it harder for attackers to deceive the system using DeepFake technologies. The multi-task learning approach within the RC-Net further improves the model's ability to extract both discriminative and representative features for authentication. However, the reliance on prerecorded data and precomputed optical flow maps may limit its performance in real-world scenarios with diverse conditions. The computational complexity could also challenge its deployment in resource-limited environments.

Continued on next page

Paper	Summary of Critical Analysis
Improving the Efficiency and Robustness of Deepfakes Detection through Precise Geometric Features	This approach utilizes facial landmarks, enhancing detection accuracy through a Landmark Calibration module and Two Stream RNN. While effective, it suffers from high computational costs, impacting real-time detection. The empirical selection of parameters may lead to inaccuracies in varying conditions, limiting practical applicability.
DF-TransFusion: Multimodal Deepfake Detection via Lip-Audio Cross-Attention and Facial Self-Attention	The use of VGG-16 for feature extraction combined with a transformer encoder for handling audio-visual tasks is a key strength. It improves efficiency by reducing frames processed, but preprocessing steps may lose valuable information. The conversion of audio from stereo to mono risks losing spatial cues. High computational complexity due to the architecture may hinder real-time applicability, suggesting a need for more efficient models.
Lips Are Lying: Spotting the Temporal Inconsistency between Audio and Visual in Lip-Syncing Deep-Fakes	This architecture uses a Global Feature Encoder based on self-attention, effective in capturing temporal dependencies for lip-sync detection. The model achieves a 90% average precision but could benefit from discussing hyperparameter impacts and including more performance metrics like F1 score. While showing robustness against common perturbations, concerns exist regarding the preprocessing of images to grayscale and audio conversion to mono, which may lead to information loss. Computational complexity raises scalability questions, suggesting a need for alternative architectures for real-time applications.

Continued on next page

Paper		Summary of Critical Analysis
Mobile DeepFake Detection through Machine Learning-based Corneal-Specular Backscattering		MobiDeep excels with its innovative use of corneal-specular backscatter images, achieving 98.7% accuracy and detection speeds under 200 milliseconds. It is a real-time mobile application that ensures privacy. However, its focus on CSB images may not withstand advanced techniques, and it requires consistent environmental parameters. The dataset lacks diversity and the comparative analysis with other methods is limited, potentially affecting its robustness.

Table 2.1: Summary of Critical Analyses on Deepfake Detection Papers

Chapter 3

Proposed Method

This chapter describes a comprehensive framework for detecting deepfake videos. Our approach leverages the accompanying audio data from these augmented videos to enhance classification performance in deepfake detection applications.

3.1 Overview

The main focus of this section is on understanding the relationship between articulators, particularly the lips, and the sounds produced during speech to effectively identify deepfakes. The key stages of our proposed method are as follows:

1. Input Video Stream
2. Pre-processing
3. Multimodal Feature Extraction and Temporal Alignment
4. Attention Mechanisms for Feature Fusion
5. Classification and Prediction

A general overview of the proposed system Figure: [3.1](#)

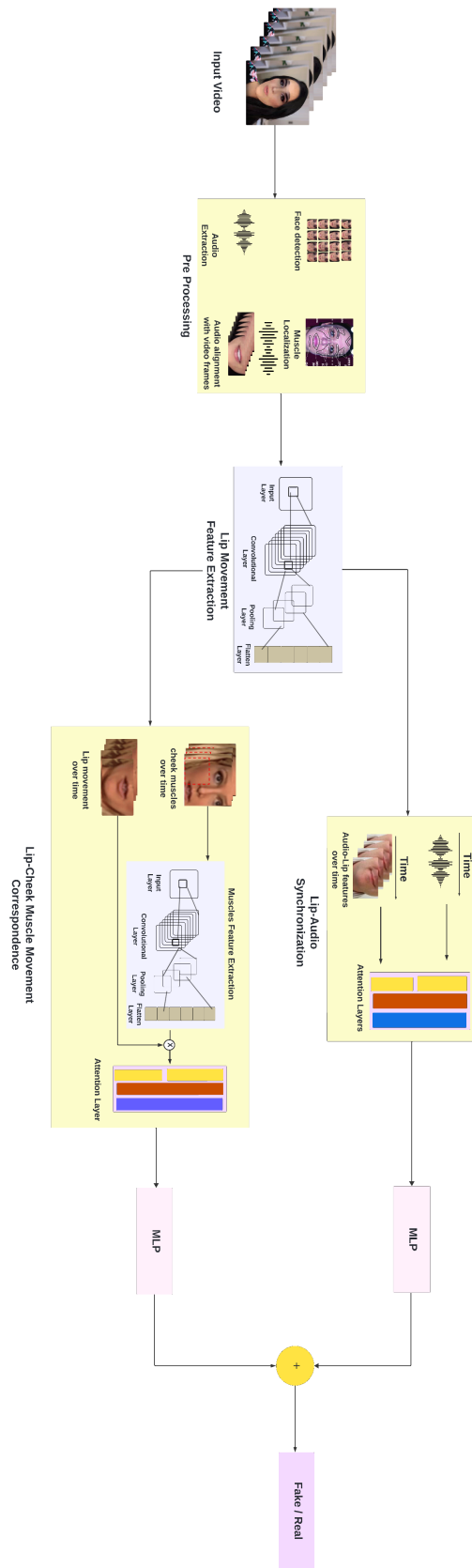


Figure 3.1: Framework of the Proposed Multi-Modal Deepfake Detection System

3.2 Methodology

Our methodology emphasizes the synchronous processing of visual and auditory information. We employ attention mechanisms to detect any discrepancies between lip movements and the corresponding audio. The detailed steps are outlined below.

3.2.1 Input Video Stream

The input consists of a video stream enriched with the appropriate audio tracks. It is crucial that the lip movements in the video correspond to the produced audio.

3.2.2 Pre-Processing

The following pre-processing procedures prepare the video and audio data for further analysis:

- **Face Detection:** Using the DLIB library[11], we identify and extract faces from each frame, isolating relevant portions of the facial structure.
- **Feature Localization:** We pinpoint key regions of interest, such as the mouth and cheeks, to focus on the areas responsible for speech production.
- **Audio Extraction:** We utilize techniques like Mel Frequency Cepstral Coefficients (MFCC)[12] and Mel Spectrogram to capture the auditory features that correspond with lip movements.
- **Frame Alignment:** We ensure that the audio and visual frames are synchronized, maintaining a temporal correlation between the features extracted from both modalities.

3.2.3 Convolutional Neural Network (CNN) for Lip Movement Feature Extraction

After pre-processing, the data is passed to a Convolutional Neural Network (CNN) to extract lip movement features. The CNN architecture comprises three convolutional layers with 32, 64, and 128 filters respectively. Each convolutional layer is followed by a max pooling layer to reduce spatial dimensions and capture essential features. The extracted features are then flattened into a one-dimensional feature vector, preparing them for subsequent analysis.

3.2.4 Attention Mechanisms for Lip-Audio Synchronization

The flattened lip movement features are fed into the Lip-Audio Synchronization block, which employs attention mechanisms to analyze both audio and lip features over time. This block consists of:

- **Self-Attention Layer:** Focuses on the temporal dynamics of lip movements, capturing dependencies across different time steps.
- **Audio Self-Attention Layer:** Extracts relevant audio features by attending to important aspects of the audio signal over time.
- **Cross-Attention Layer:** Integrates lip and audio features, enhancing the synchronization between lip movements and corresponding audio signals.

These attention layers work together to identify inconsistencies between lip movements and audio, which are indicative of Deepfake manipulations.

3.2.5 CNN and Cross-Attention Layers for Lip-Cheek Muscle Movement Correspondence

Parallel to lip-audio synchronization, the lip movement features are also processed in the Lip-Cheek Muscle Movement Correspondence block. This block focuses on analyzing cheek muscle movements over time through:

- **Cheek Muscle CNN:** A dedicated CNN extracts detailed features related to cheek muscle movements from the video frames.
- **Cross-Attention Layer:** Combines the extracted cheek muscle features with the lip movement features, allowing the model to correlate movements in both regions.

This integration helps in detecting subtle inconsistencies between lip and cheek movements, further improving Deepfake detection accuracy.

3.2.6 Multi-Layer Perceptron (MLP) for Classification and Feature Fusion

After processing through the attention mechanisms, the synchronized features from the Lip-Audio Synchronization block are passed to a Multi-Layer Perceptron (MLP) for initial classification. Simultaneously, the output from the Lip-Cheek Muscle Movement Correspondence

block is fed into a separate MLP. The outputs from both MLPs are then combined through a feature fusion process, which aggregates the predictions from the different modules. This fused output is used to generate the final classification decision, determining whether the input video is real or fake.

References

- [1] Haojie Wu, Pan Hui, and Pengyuan Zhou. Deepfake in the metaverse: An outlook survey. *arXiv preprint arXiv:2306.07011*, 2023.
- [2] KR Prajwal, Rudrabha Mukhopadhyay, Vinay P Namboodiri, and CV Jawahar. A lip sync expert is all you need for speech to lip generation in the wild. In *Proceedings of the 28th ACM international conference on multimedia*, pages 484–492, 2020.
- [3] Ting-Chun Wang, Arun Mallya, and Ming-Yu Liu. One-shot free-view neural talking-head synthesis for video conferencing. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10039–10049, 2021.
- [4] Trisha Mittal, Uttaran Bhattacharya, Rohan Chandra, Aniket Bera, and Dinesh Manocha. Emotions don’t lie: An audio-visual deepfake detection method using affective cues. In *Proceedings of the 28th ACM international conference on multimedia*, pages 2823–2832, 2020.
- [5] Xin Liao, Yumei Wang, Tianyi Wang, Juan Hu, and Xiaoshuai Wu. Famm: facial muscle motions for detecting compressed deepfake videos over social networks. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(12):7236–7251, 2023.
- [6] Zekun Sun, Yujie Han, Zeyu Hua, Na Ruan, and Weijia Jia. Improving the efficiency and robustness of deepfakes detection through precise geometric features. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3608–3617, 2021.
- [7] Zekun Sun, Yujie Han, Zeyu Hua, Na Ruan, and Weijia Jia. Improving the efficiency and robustness of deepfakes detection through precise geometric features. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3609–3618, 2021.
- [8] Aaditya Kharel, Manas Paranjape, and Aniket Bera. Df-transfusion: Multimodal deepfake detection via lip-audio cross-attention and facial self-attention. *arXiv preprint arXiv:2309.06511*, 2023.

- [9] Weifeng Liu, Tianyi She, Jiawei Liu, Run Wang, Dongyu Yao, and Ziyu Liang. Lips are lying: Spotting the temporal inconsistency between audio and visual in lip-syncing deepfakes. *arXiv preprint arXiv:2401.15668*, 2024.
- [10] Muhammad Mohzary, Khalid J Almalki, Baek-Young Choi, and Sejun Song. Mobideep: Mobile deepfake detection through machine learning-based corneal-specular backscattering. In *2023 IEEE 20th Consumer Communications & Networking Conference (CCNC)*, pages 1104–1109. IEEE, 2023.
- [11] Davis E. King. Dlib c++ library. <http://dlib.net/>, 2024. Accessed: 2024-10-05.
- [12] Md Afzal Hossan, Sheeraz Memon, and Mark A Gregory. A novel approach for mfcc feature extraction. In *2010 4th international conference on signal processing and communication systems*, pages 1–5. IEEE, 2010.